

Ali Nikkhah

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TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, JAX, TensorFlow, Hugging Face (Transformers), Time-Series Transformers, ONNX Runtime
Agentic AI & Advanced RAG: LangGraph, LangChain (LCEL), LlamaIndex Workflows, CrewAI, AutoGen, Hybrid Search, HyDE, Graph RAG, KAG
Vision & Audio Systems: OpenCV, BLIP, Vision Transformers (ViT), T5, Whisper, wav2vec2.0, Librosa, ESPnet
Data Engineering & Lakes: Apache Spark (PySpark), Trino, ClickHouse, Apache Airflow, PeerDB, Apache Kafka, RabbitMQ, Parquet
MLOps & Infrastructure: Docker, Kubernetes, Ray, Kubeflow, Triton Inference Server, TensorFlow Serving, TensorRT-LLM, TF Lite
DevOps & Automation: Terraform, GitHub Actions, Jenkins, GitLab CI
Monitoring, Governance & BI: Prometheus, Grafana, ELK Stack, Metabase, Weights & Biases, MLflow, DVC

PROFESSIONAL EXPERIENCE

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Scientist & AI Engineer

Aug 2025 - Mar 2026

- Engineered an enterprise telephony voice intelligence pipeline, utilizing transformer-based speech emotion recognition architectures (Whisper, wav2vec2.0) to capture, transcribe, and analyze live audio data streams.
- Designed multimodal data fusion primitives to synthesize acoustic voice signals, user interaction metadata, and textual transaction logs into unified embedding representations for downstream predictive classification models.
- Aligned historical time-series pricing vectors with high-frequency market sentiment news text embeddings, deploying deep sequence models to map cross-modality pricing correlations and market turbulence effects.
- Built asynchronous audio streaming workers utilizing Python's Asyncio library to process parallel incoming voice payloads without blocking main thread API endpoints.
- Optimized downstream model latency by compiling acoustic feature extraction blocks directly into TorchScript, accelerating runtime execution across automated ticketing clusters.
- Configured continuous validation frameworks to monitor acoustic signal-to-noise ratios, dynamically adjusting transcription thresholds under poor telecommunication connectivity conditions.
- Supervised the design of automated pipeline instrumentation to log acoustic distribution shifts, feeding visual metrics into custom deployment dashboards.

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 - Aug 2025

- Led the development of multimodal retrieval systems, coordinating the deployment of vision-language architectures to index, align, and fetch non-homogenous asset modalities (images, diagrams, and text product profiles).
- Optimized vision-model execution matrices on internal serving infrastructure using Triton Inference Server, setting up dynamic request batching to handle multi-stream image tokenization profiles.
- Integrated Vision Transformers (ViT) and convolutional feature extractors into core search pipelines, establishing shared vector spaces to enable text-to-image semantic searching.
- Compiled deep vision and sequence networks into hardware-optimized runtimes utilizing ONNX Runtime and TensorRT, reducing downstream inference latencies across the model registry layer.
- Implemented static weight quantization architectures on visual feature extraction models to minimize computational costs across corporate host environments.
- Programmed high-throughput image preprocessing and spatial augmentation workers, utilizing optimized libraries to decouple raw upload streams from critical inference execution blocks.
- Conducted comprehensive performance profiling evaluations on multimodal alignment frameworks, optimizing matrix operations to prevent out-of-memory (OOM) failures under heavy traffic bursts.

HomaCloud

Tehran, Iran

MLOps Engineer

May 2021 - Feb 2022

- Designed and deployed scalable MLOps pipelines on Kubernetes clusters to automate the training and distribution of deep convolutional networks and spatial-temporal processing models.

- Automated containerized CI/CD pipelines integrating Docker, ArgoCD, and GitHub Actions, streamlining runtime profiling for vision libraries like OpenCV across distributed inference nodes.
- Conducted extensive profiling evaluations on deep vision backbones within PyTorch and TensorFlow, identifying processing bottlenecks in spatial augmentation and temporal frame sampling routines.
- Configured automated infrastructure alerting triggers inside Prometheus to track graphic processing unit (GPU) memory capacities during heavy computer vision deployment jobs.

Mapna-MD2

Software Engineer

Tehran, Iran

Nov 2022 - Jan 2023

- Developed backend microservices to ingest high-frequency industrial IoT camera and sensor feeds, building structural streaming interfaces to pass telemetry data into processing queues.
- Designed and implemented interactive web data dashboards and telemetry visualization views, mapping high-velocity backend visual log arrays to frontend application states cleanly.
- Optimized database read routines for high-dimensional historical monitoring metrics, accelerating tracking canvas generation speeds inside engineering client interfaces.
- Programmed custom data formatting utility scripts to serialize spatial-temporal sensor payloads into lightweight, compressed data buffers.

SELECTED PROJECTS & RESEARCH

Advanced Analytics Australia

Computer Vision Engineer

Remote Contract

Oct 2024 - Apr 2025

- Engineered edge and server-side video analytics pipelines on AXIS camera systems to automate health, safety, and compliance auditing, including handwash tracking, protective gear detection, and restricted zone entry monitoring.
- Optimized deep vision network deployment across resource-constrained edge computing hardware by utilizing static model quantization, layer compilation, and concurrent container hosting configurations via Triton Inference Server and TensorFlow Lite.
- Architected a high-throughput spatial metadata ingestion broker using Apache Kafka combined with optimized GPU memory caching routines to handle concurrent high-frame-rate tracking streams and dynamic sensor calibration.

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering - Digital Systems (GPA: 3.65, Major: 3.91)

Tehran, Iran

Sept 2020 - Jul 2024